NONCOMMUTATIVE SPACE AND THE ORIGIN OF THE QUANTUM WAVEFUNCTION

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Abstract. These are concise personal notes for a very gentle introduction to noncommutative geometry as it appears in physics from the mathematical point of view, edited to be readable by others. Their main goal is to explain the analogy that "wavefunctions are quantum points".

1. Abstract C^*-algebras

Given any (locally compact Hausdorff) topological space X, we can define $C_0(X)$, the algebra of $\mathbb{C}$-valued functions which vanish at ∞.[1] This algebra is very nice. In particular, it is a *C^*-algebra*.

Definition 1.1. A *C^*-algebra* is a Banach space over $\mathbb{C}$ together with

- a multiplication operation that turns it into a (not necessarily unital) associative algebra over $\mathbb{C}$
- a norm $||\cdot||$ satisfying $||xy|| \leq ||x||\,||y||$
- an involution $x \mapsto x^*$ satisfying
  - $(x+y)^* = x^* + y^*$ and $(xy)^* = y^*x^*$
  - $(\lambda x)^* = \bar{\lambda} x^*$
  - $||xx^*|| = ||x||\,||x^*||$

Using the sup norm and with complex conjugation as the involution, $C_0(X)$ becomes a commutative C^*-algebra. Surprisingly, it's possible to go the other way and reconstruct X entirely from $C_0(X)$! Given any C^*-algebra A, we can define the *spectrum* Φ_A.

Definition 1.2. The *spectrum* of a C^*-algebra A is the set of nonzero algebra homomorphisms from A to $\mathbb{C}$.

We can equip the spectrum with the weak-* topology, which makes it locally compact and Hausdorff. If A is an algebra of continuous functions on a space, then pointwise evaluation gives an element of the spectrum for every element in the space. But we're not assuming that A is of this form! Nevertheless,

Theorem 1.3. *(Commutative Gelfand-Naimark) For any C^*-algebra A, there is the isometric isomorphism*

$$A \simeq C_0(\Phi_A)$$

Corollary 1.4. *$C_0(X) \simeq C_0(Y)$ iff X is homeomorphic to Y.*

So we can study spaces solely by studying their commutative C^*-algebra of functions! The goal of noncommutative geometry is to generalize this study to noncommutative C^*-algebras and thereby construct "noncommutative" generalizations of spaces.

[1] For arbitrary locally compact spaces, these are functions which, $\forall \epsilon > 0$, are bounded by ϵ outside of some compact set K_ϵ.

2. The fuzzy sphere

The algebra of operators in quantum mechanics is noncommutative. In particular, we have the *canonical commutation relation*

$$[\hat{x}, \hat{p}] = i\hbar.$$

The ultimate goal of mathematical physicists is to describe the "noncommutative geometry" of observables, where we now interpret phase space as a noncommutative space with corresponding noncommuting coordinate functions. However, this algebra as stated is incredibly poorly behaved (it can't even be given a C^*-algebra structure), so complicated workarounds are needed. Because of this, instead of working with a physically relevant algebra, we are going to describe a simpler model and hope that we can still glean some useful information.

Consider the $\mathfrak{su}(2)$ Lie algebra, with generators J_1, J_2, J_3 and bracket

$$[J_i, J_j] = i\epsilon_{ijk}J_k$$

using the Levi-Civita symbol ϵ_{ijk}. (This is a hybrid between the mathematics and physics conventions, with Hemitian generators but no factor of 2.) Pick an N-dimensional irrep of $\mathfrak{su}(2)$, and hence a representation of its universal enveloping algebra. This algebra has a Casimir element

$$J_1^2 + J_2^2 + J_3^2 = \frac{N^2 - 1}{4}$$

We can (assuming $N > 1$) define new operators

$$\hat{x}_i = \frac{2R}{\sqrt{N^2 - 1}} J_i$$

with $R > 0$ constant. These new operators satisfy.

$$\hat{x}_1^2 + \hat{x}_2^2 + \hat{x}_3^2 = R^2$$

which is exactly the equation for a sphere of radius R. We can therefore interpret the $\hat{x}_i$ as noncommutative coordinates on the *fuzzy sphere*, with commutation relation

$$[\hat{x}_i, \hat{x}_j] = \frac{2R}{\sqrt{N^2 - 1}} i\epsilon_{ijk}\hat{x}_k$$

Observe that in the $N \to \infty$ limit the coordinates commute and we recover the classical sphere.

We can also imagine functions on the fuzzy sphere. Generically, polynomials in the $\hat{x}_i$ will be $N \times N$ matrices. By Burnside's theorem on matrix algebras, every $N \times N$ matrix can be realized this way. Therefore, it makes sense to call $\mathrm{Mat}_{N\times N}$ the "algebra of functions" over the fuzzy sphere. As we had hoped, $\mathrm{Mat}_{N\times N}$ is a noncommutative C^*-algebra with the Hermitian conjugate as its involution.

Note that unlike the classical space of continuous functions on the sphere, $\mathrm{Mat}_{N\times N}$ is finite-dimensional. This has made the fuzzy sphere a very valuable tool for approximating complicated computations on S^2, for example as appearing in quantum field theory.

3. States on the fuzzy sphere

We want to recover the notion of a "point" on the fuzzy sphere. For this, we go back to the commutative case. From the perspective of the C^*-algebra, points "are" just their evaluation functionals. Specifically, point act as positive linear functionals on the algebra.

Definition 3.1. A linear functional on a C^*-algebra A is *positive* if it sends elements of the form a^*a to positive real numbers

We have by Riesz representation that the positive linear functionals on the space of continuous functions are given by positive Radon measures on X.

$$\langle f, \mu \rangle \mapsto \int_X f d\mu$$

The evaluation functionals are then just Dirac measures, but we can in principle consider any Radon measure as an evaluation of sorts. In particular, if we let μ be a probability distribution

$$\int_X d\mu = 1$$

then this is just an expectation functional. These probability measures define a convex subset of the topological vector space of Radon measures. What makes points special in this framework is that they are the *extreme points* of this convex set (they are not a convex combination of other elements of the set).

We can generalize this definition by defining a *state*.

Definition 3.2. A *state* of a C^*-algebra is a positive linear functional of norm 1.

Radon probability measures are the states of commutative C^*-algebras, but note that the above definition doesn't rely on the C^*-algebra being commutative! Consider the fuzzy sphere from earlier. By Riesz representation, we know all linear functionals are of the form

$$A \mapsto \mathrm{Tr}(A\rho)$$

In order for ρ to be a state, we need to guarantee that it is positive semidefinite and that $\mathrm{Tr}(\rho) = 1$. These are exactly the same conditions as ρ being a *density matrix*. The functionals then have the same form as expectations of density matrices in quantum mechanics, just as the the commutative case was an expectation in classical mechanics.

States always define a convex set in the topological vector space of linear functionals (with the weak-* topology). We can therefore always generalize the notion of "point" by the Krein-Milman theorem.

Theorem 3.3. *(Krein-Milman) A compact convex subset of a Hausdorff locally convex topological vector space is equal to the closed convex hull of its extreme points.*

Explicitly, every density matrix is given by a convex combination of rank 1 projection operators

$$P = |\psi\rangle \langle\psi|$$

where $|\psi\rangle$ is a vector, exactly like the quantum wavefunction! We therefore see that P is exactly analogous to pointwise evaluation, and the wavefunction is exactly the noncommutative analogue of a point.

4. The general case

It might seem like our above observations relied on being able to write our noncommutative function space as an algebra of operators. However, this turns out to be a general phenomenon.

Theorem 4.1. *(Gelfand-Naimark) Any C^*-algebra is isometrically isomorphic to a C^*-subalgebra of bounded operators on a Hilbert space.*

So all of the observations about density matrices and wavefunctions apply to *any* C^*-algebra, including physically relevant ones! Things that are typically seen as distinctly quantum-mechanical, including

- observables being operators on a Hilbert space
- measurement outcomes being determined by a wavefunction

are actually just a consequence of the structure of C^*-algebras.